

UKAS QUALITY MANAGEMENT 005

SYSTEM CERTIFICATION ISO 9001:2008 SGS

ACROW

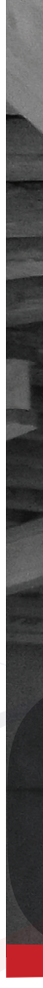
Formwork Technology

Acrow flex System

ACROW MISR Provides Full Site Service Based On More Than 35 Years Experience In The Field

ACROW

Formwork Technology





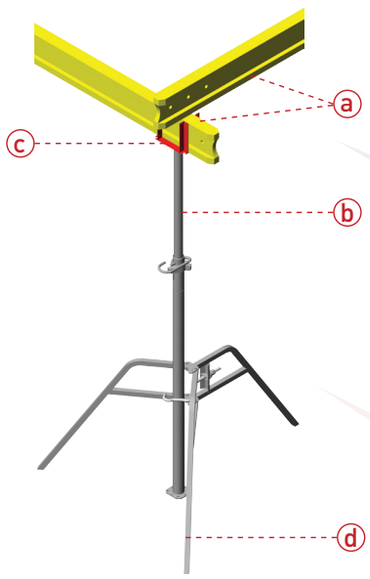
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ACROW
Formwork Technology

Intoduction

•The Acrowflex is a slab formwork system that could be adapted to any floor plan.
•It consists mainly of H20 wooden beams that work as main and secondary girders.
•The beams are supported by props.
•Acrowflex is the cost effective solution for slabs of thickness up to 50 cm.
•Acrowflex is mainly when shoring height is less than 4.5 m.
•Few system components, only H20 girders, Props, stirrup heads and Prop legs.



a- H20 beams as main and secondary girders.
b- A.E.P or Alu props for shoring.
c- Stirrup head.
d- Prop leg.



Acrow European Props(A.E.P) for shoring

Acrow Slab Props

AEP - Acrow European Prop

•The Prop AEP is designed to be used as a support element for Acrow horizontal formwork.

•The AEP is a hot dip galvanized Prop having high durability.

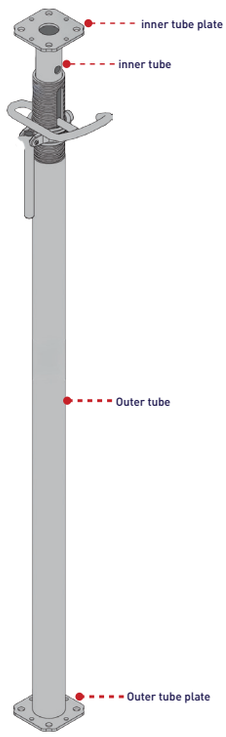
•Using the AEP (Acrow European Prop) gives high load capacity.

•Permanent preventaton against unintentional disengagement.

•The inner and outer parts of a prop shall be prevented permanently from being separated except by intentional action.

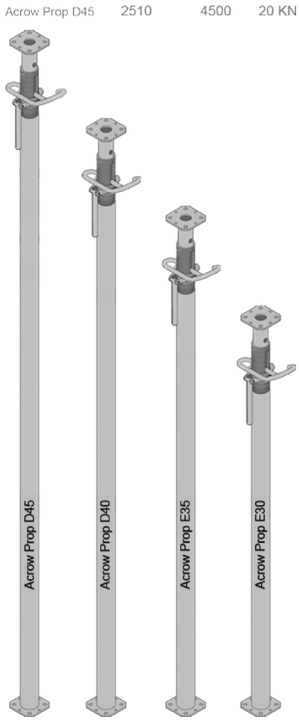
•Anti hand trap. When the prop is fuly closed without the pin inserted, there is 100 mm clearance between the endmost part of the outer tube and the inner tube.





Extension (cm)	G 300 reg. DIN	E 30/300	G 350 reg. DIN	E 30/350	G 410 reg. DIN	D 20/400 nach EN	G 490 reg. DIN	D 20/450 nach EN
150								
160								
180	35.0	30 kN						
200	35.0		35.0	30 kN				
220	35.0		35.0		35.0	20 kN		
240	31.3		35.0		35.0			
250	28.8		33.6		35.0		35.0	20 kN
260	26.6		31.1		35.0		35.0	
280	23.0		26.8		31.4		35.0	
300	20.0		23.3		27.3		30.5	
320			20.5		24.0		28.7	
340			18.2		21.3		25.4	
350			17.1		20.1		24.3	
360					19.0		22.7	
380					17.0		20.4	
400					15.4		18.4	
420							16.7	
450							14.5	
460								
480								
named reg. EN 1065	C30	E30	C35	E35	C40	D40	C45	D45

Product name	Min. Extension mm	Max. Extension mm	Max. Load KN
Acrow Prop E30	1740	3000	30 KN
Acrow Prop E35	1990	3500	30 KN
Acrow Prop D40	2240	4000	20 KN
Acrow Prop D45	2510	4500	20 KN



Acrow slab props

Alu Props:
The ALU-PROP post shores are made of aluminium which makes the props light and easy in handling.

Four different sizes of ALU-PROPS are available, ALP 300,400,500 and 600, the ALP300 can be extended from 1.708 m to 3.0 m but weights only 16.0 Kg.

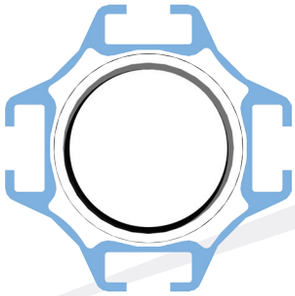
Product advantages:
•Height adjustment without using pins,the inner tube (a) is threaded not like the conventional steel props fitted.

•The nut and the key (b) make safety stops as the inner tube can not slide out accidently during transportation.

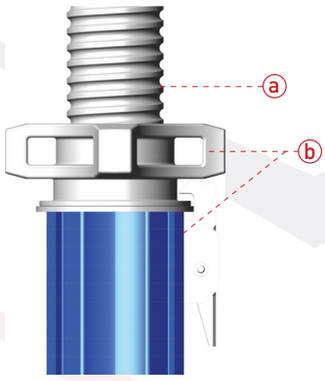
•Economic as it can be used as shoring towers by bracing it with the alu-frames or as single props.

•Alu-Props could be braced using swivel couplers and tubes.

•High load capacity: 60 kN



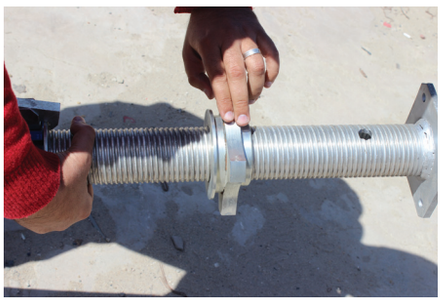
Alu prop profile







adjusting the height is easy, just release the inner tube, then adjust with the nut.



Threaded inner tube with the nut makes accurate adjustment.



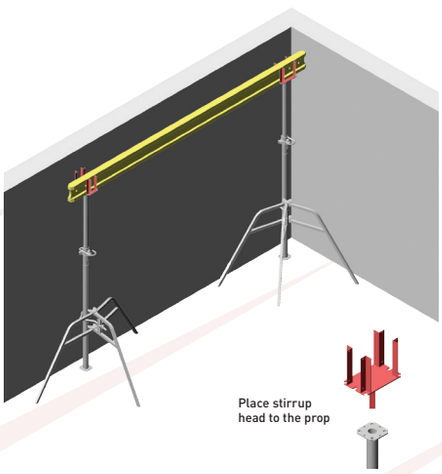
ALU PROP with ALU DECK panels, fitted with Drophead



Bracing Shoringtowers with Alu Frames

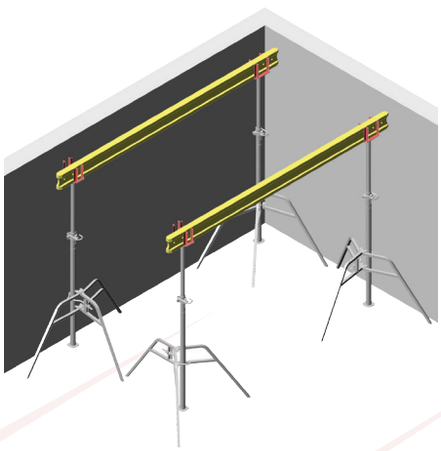
Erection

1



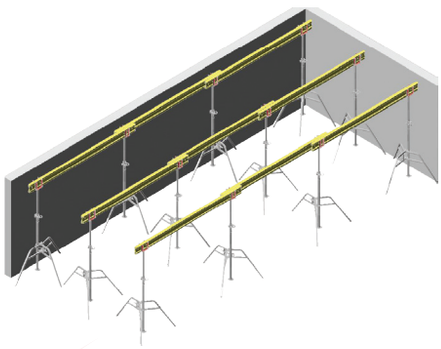
•Attach the Prop leg to the props in order to be stable as in picture 1.
•Install the H20 timber beam (size according to design drawings).

2



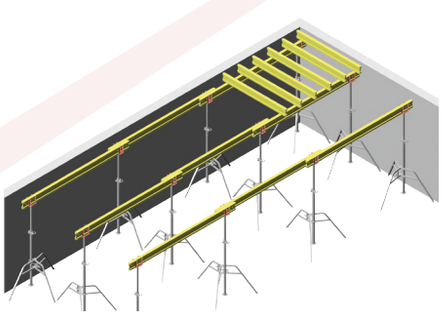
•Erect the second row of timber beams and props as in picture 2 according to the design drawings.

3



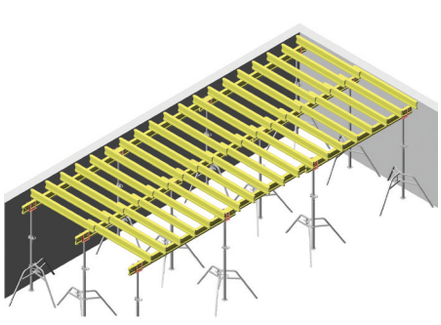
•Continue erecting the rest of the main beams with the main props.

4



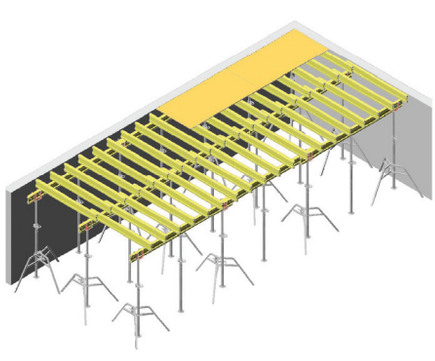
•Install secondary H20 beams as in picture 4.

5



•Installing secondary H20 beams.

6



•Attach the intermediate props between the main props according to the design spacing.
•The plywood is ready to be installed.

Design Tables

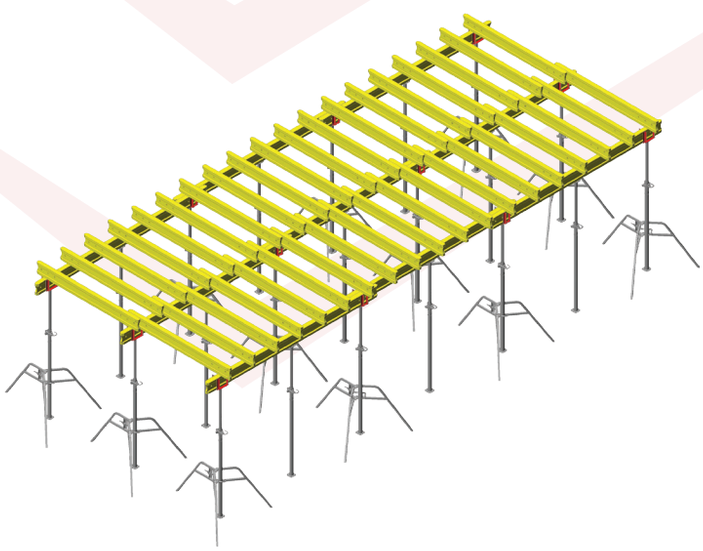
Secondary Girder: H20
Main Girder: H20

Slab Thickness (m)		0.10			0.12			0.14			0.16			0.18			0.20		
Load q (kN/m^2)		4.50			5.00			5.50			6.10			6.60			7.10		
Sec. Girder Spacing (m)		0.75	0.625	0.50	0.75	0.625	0.50	0.75	0.625	0.50	0.75	0.625	0.50	0.75	0.625	0.50	0.75	0.625	0.50
Prop Spacing (m)	0.60	3.10	3.30	3.55	2.94	3.13	3.37	2.81	2.99	3.22	2.70	2.87	3.09	2.60	2.77	2.98	2.52	2.68	2.89
		8.40	8.90	9.60	8.90	9.40	10.10	9.30	9.90	10.70	9.80	10.00	11.20	10.30	10.90	11.80	10.70	11.40	12.30
	0.90	3.10	3.30	3.55	2.94	3.13	3.37	2.81	2.99	3.22	2.70	2.87	3.09	2.60	2.77	2.98	2.52	2.68	2.89
		12.60	13.40	14.40	13.30	14.10	15.20	14.00	14.90	16.00	14.70	15.60	16.90	15.40	16.40	17.70	16.10	17.10	18.40
	1.20	3.10	3.30	3.55	2.94	3.13	3.37	2.81	2.99	3.22	2.70	2.87	3.09	2.60	2.77	2.79	2.52	2.58	2.58
		16.80	17.80	19.20	17.70	18.80	20.30	18.70	19.90	21.40	19.60	20.90	20.60	21.80	22.00	21.50	22.00	22.00	22.00
	1.50	3.10	3.26	3.26	2.92	2.92	2.92	2.65	2.65	2.65	2.42	2.42	2.42	2.23	2.23	2.23	2.07	2.07	2.07
		21.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00
	1.80	2.50	2.50	2.50	2.24	2.24	2.24	2.03	2.03	2.03	1.86	1.86	1.86	1.71	1.71	1.71	1.59	1.59	1.59
		22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00
	2.10	1.91	1.91	1.91	1.71	1.71	1.71	1.55	1.55	1.55	1.42	1.42	1.42	1.30	1.30	1.30	1.21	1.21	1.21
		22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00
	2.40	1.54	1.54	1.54	1.38	1.38	1.38	1.25	1.25	1.25	1.15	1.15	1.15	1.06	1.06	1.06	0.98	0.98	0.98
		22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00

Slab Thickness (m)		0.22			0.24			0.26			0.28			0.30			0.35	
Load q (kN/m^2)		7.60			8.10			8.70			9.20			9.80			11.30	
Sec. Girder Spacing (m)		0.75	0.625	0.50	0.625	0.50	0.40	0.625	0.50	0.40	0.625	0.50	0.40	0.625	0.50	0.40	0.50	0.40
Prop Spacing (m)	0.60	2.45	2.60	2.80	2.53	2.73	2.94	2.47	2.66	2.86	2.41	2.60	2.80	2.36	2.54	2.74	2.42	2.61
		11.20	11.90	12.80	12.40	13.30	14.30	12.80	13.80	14.90	13.30	14.30	15.40	13.80	14.90	16.00	16.40	17.70
	0.90	2.45	2.60	2.80	2.53	2.73	2.94	2.47	2.66	2.82	2.41	2.60	2.66	2.36	2.50	2.50	2.16	2.16
		2.00	17.80	19.20	18.50	20.00	21.50	19.20	20.70	22.00	19.90	21.50	22.00	20.70	22.00	22.00	22.00	22.00
	1.20	2.41	2.41	2.41	2.25	2.25	2.25	2.12	2.12	2.12	2.00	2.00	2.00	1.88	1.88	1.88	1.62	1.62
		22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00
	1.50	1.92	1.92	1.92	1.80	1.80	1.80	1.69	1.69	1.69	1.60	1.60	1.60	1.50	1.50	1.50	1.30	1.30
		22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00
	1.80	1.48	1.48	1.48	1.38	1.38	1.38	1.30	1.30	1.30	1.23	1.23	1.23	1.15	1.15	1.15	1.00	1.00
		22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00
	2.10	1.13	1.13	1.13	1.05	1.05	1.05	0.99	0.99	0.99	0.93	0.93	0.93	0.88	0.88	0.88	0.76	0.76
		22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00
	2.40	0.91	0.91	0.91	0.85	0.85	0.85	0.80	0.80	0.80	0.76	0.76	0.76	0.71	0.71	0.71	0.61	0.61
		22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00	22.00

Slab Thickness (m)		0.40		0.45		0.50	
Load q (kN/m^2)		12.90		14.40		16.00	
Sec. Girder Spacing (m)		0.50	0.40	0.50	0.40	0.50	0.40
Prop Spacing (m)	0.60	2.32	2.50	2.23	2.40	2.16	2.29
		17.90	19.30	19.30	20.80	20.70	22.00
	0.90	1.90	1.90	1.69	1.69	1.53	1.53
		22.00	22.00	22.00	22.00	22.00	22.00
	1.20	1.42	1.42	1.27	1.27	1.15	1.15
		22.00	22.00	22.00	22.00	22.00	22.00
	1.50	1.14	1.14	1.02	1.02	0.92	0.92
		22.00	22.00	22.00	22.00	22.00	22.00
	1.80	0.87	0.87	0.78	0.78	0.70	0.70
		22.00	22.00	22.00	22.00	22.00	22.00
	2.10	0.67	0.67	0.59	0.59	0.54	0.54
		22.00	22.00	22.00	22.00	22.00	22.00
	2.40	0.54	0.54	0.48	0.48	0.43	0.43
		22.00	22.00	22.00	22.00	22.00	22.00

	Main girder spacing (m)
	Sec. girders spacing(m)



Dead load g = 0.40 kn/m2
Concrete load b = 25 kn/m3 x d (m)
Live load p = 0.20 x b
$1.5 \leq p \leq 5.0$ kN/m2
Total load q = g + b + p

The deflection has been limited to l/500. Sec.
girders assumed as single span.

Design Tables

Secondary Girder: H20
Main Girder: 2 x H20

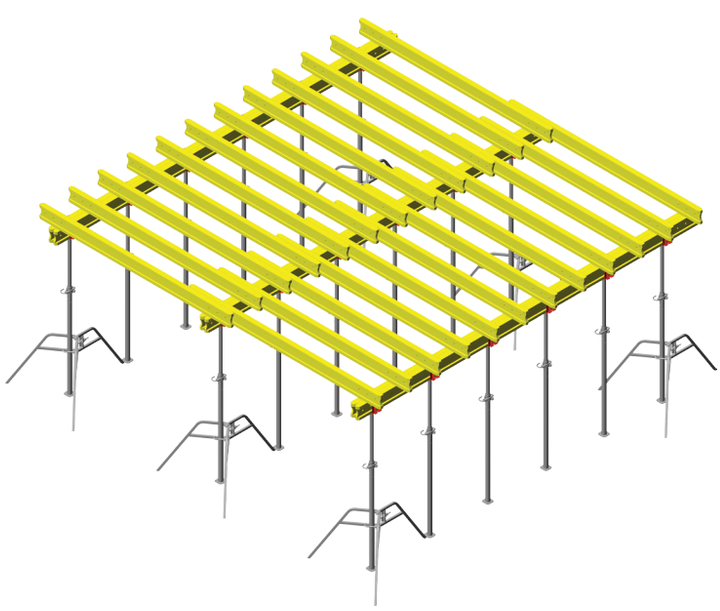
Slab Thickness (m)		0.22			0.24			0.25			0.26			0.28		
Load q (kN/m^2)		7.60			8.10			8.40			8.70			9.20		
Sec. Girder Spacing (m)		0.75	0.625	0.50	0.625	0.50	0.40	0.625	0.50	0.40	0.625	0.50	0.40	0.625	0.50	0.40
Prop Spacing (m)	0.50	2.45	2.60	2.80	2.53	2.73	2.94	2.50	2.69	2.90	2.47	2.66	2.86	2.41	2.60	2.80
		9.30	9.90	10.70	10.30	11.10	12.00	10.50	11.30	12.18	10.74	11.57	12.44	11.09	11.96	12.88
	0.75	2.45	2.60	2.80	2.53	2.73	2.94	2.50	2.69	2.90	2.47	2.66	2.86	2.41	2.60	2.80
		14.00	14.90	16.00	14.50	15.60	16.80	15.80	16.90	18.27	16.12	17.36	18.66	16.63	17.94	19.32
	1.00	2.45	2.60	2.80	2.53	2.73	2.94	2.50	2.69	2.90	2.47	2.66	2.86	2.41	2.60	2.80
		18.70	19.80	21.30	20.60	22.20	23.90	21.00	22.60	24.36	21.49	23.14	24.88	22.17	23.92	25.76
	1.25	2.45	2.60	2.80	2.53	2.73	2.94	2.50	2.69	2.90	2.47	2.66	2.86	2.41	2.60	2.80
		23.30	24.80	26.70	25.70	27.80	29.90	26.30	28.20	30.45	26.86	28.93	31.10	27.72	29.90	32.20
	1.50	2.45	2.60	2.80	2.53	2.73	2.94	2.50	2.69	2.90	2.47	2.66	2.86	2.41	2.60	2.80
		28.00	29.70	32.00	30.90	33.30	35.90	31.50	33.90	36.54	32.23	34.71	37.32	33.26	35.88	38.64
	1.75	2.45	2.60	2.80	2.53	2.73	2.94	2.50	2.69	2.90	2.47	2.66	2.86	2.41	2.60	2.74
		32.70	34.70	37.30	36.00	38.90	41.90	36.80	39.54	42.63	37.61	40.50	43.54	38.80	41.86	44.11

Slab Thickness (m)		0.30			0.35		0.40		0.45		0.50	
Load q (kN/m^2)		9.80			11.30		12.90		14.40		16.00	
Sec. Girder Spacing (m)		0.625	0.50	0.40	0.50	0.40	0.50	0.40	0.50	0.40	0.50	0.40
Prop Spacing (m)	0.50	2.36	2.54	2.74	2.43	2.62	2.32	2.50	2.24	2.41	2.16	2.32
		11.56	12.45	13.43	11.91	12.84	14.96	16.13	16.13	17.35	17.28	18.56
	0.75	2.36	2.54	2.74	2.43	2.62	2.32	2.50	2.24	2.41	2.16	2.32
		17.35	18.67	20.14	20.59	22.20	22.45	24.19	24.19	26.03	25.92	27.84
	1.00	2.36	2.54	2.74	2.43	2.62	2.32	2.50	2.24	2.41	2.16	2.32
		23.13	24.89	26.85	27.46	29.61	29.93	32.25	32.26	34.70	34.56	37.12
	1.25	2.36	2.54	2.74	2.43	2.62	2.32	2.50	2.24	2.41	2.16	2.32
		28.91	31.12	33.57	34.32	37.01	37.41	40.31	40.32	43.38	43.20	46.40
	1.50	2.36	2.54	2.74	2.43	2.59	2.28	2.28	2.03	2.03	1.83	1.83
		34.69	31.12	33.57	41.19	43.90	44.12	44.12	43.85	43.85	43.92	43.92
	1.75	2.36	2.54	2.74	2.22	2.22	1.95	1.95	1.74	1.74	1.57	1.57
		40.47	31.12	33.57	43.90	43.90	44.02	44.02	43.85	43.85	43.96	43.96

	Main girder spacing (m)
	Sec. girders spacing(m)

Dead load g = 0.40 kn/m2
Concrete load b = 25 kn/m3 x d (m)
Live load p = 0.20 x b
$1.5 \leq p \leq 5.0$ kN/m2
Total load q = g + b + p

the deflection has been limited to l/500. Sec.
girders assumed as single span.



Acrowflex early striking (AES)

•The AES is an integrated system for early striking.

•The AES allows the stripping operation to be pre-defined.

•The AES drophead has a quick lowering function for minimizing damage when striking. It also keeps props stables during formwork removal.

•The drophead principle allows early striking after 1 day depending on the slab thickness and the concrete strength.

•Props on the edges or infill areas can be dismantled and used for the next cycle,thus reducing shoring quantities.

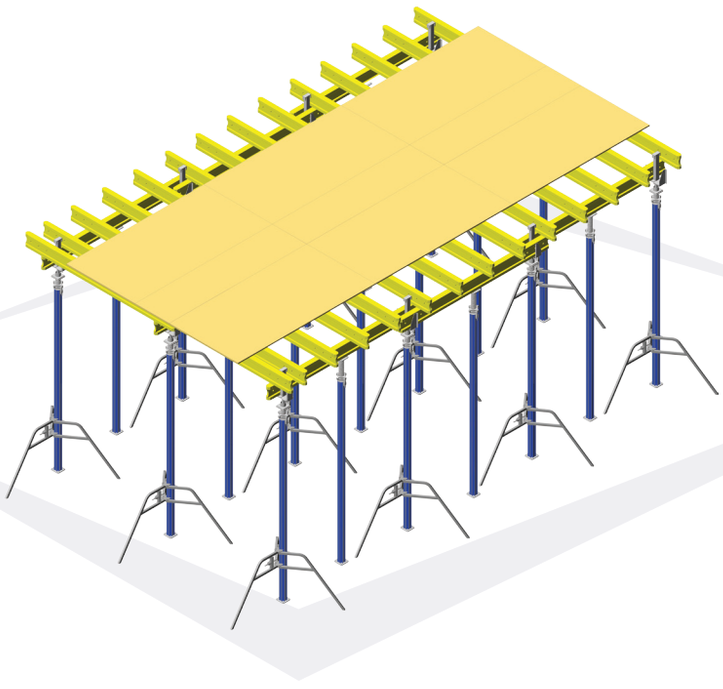
•When the floor props are used for temporary back propping, their permitted load-bearing capacity is 10kN higher.

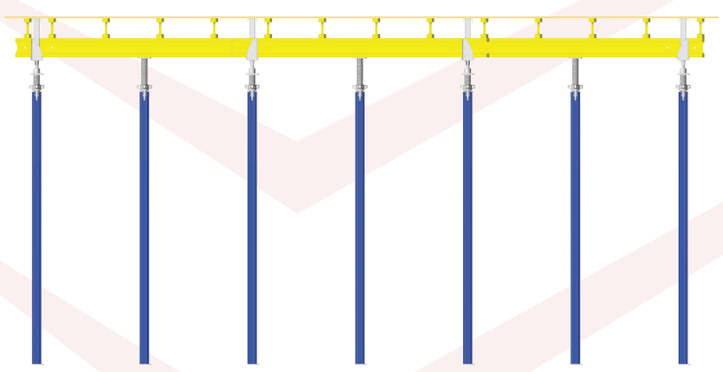
Important note:

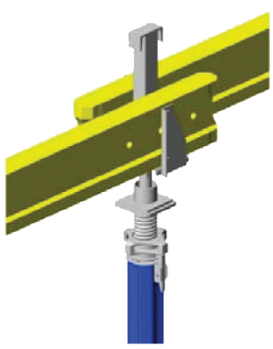
•Floor props fitted with the drophead must not be extended to their full length.

•Permitted floor-prop extension length is less 50 cm than the floor prop maximum length.

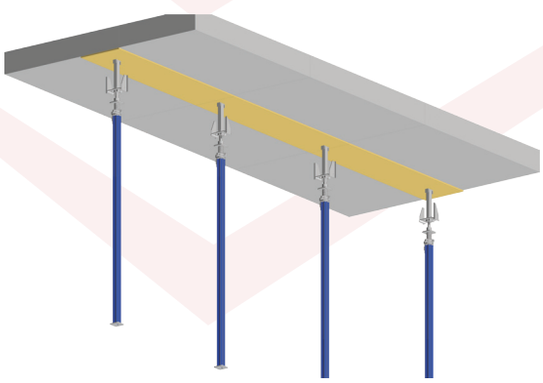
Example: floor prop D45 is allowed to be extended to max. 400 cm.







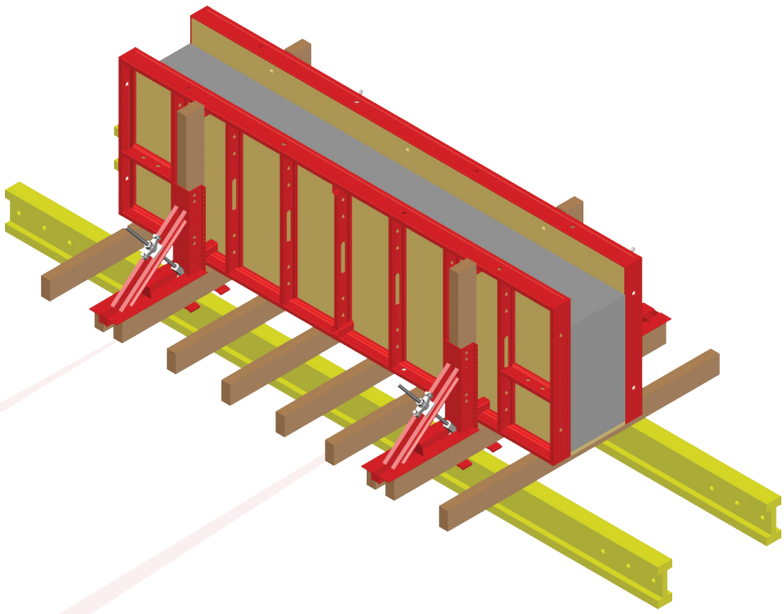
Drophead with H20 as main girders

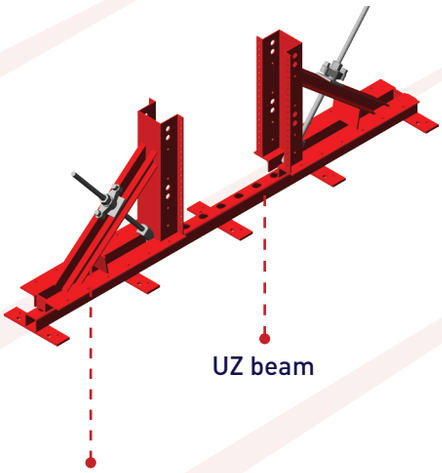


After removal of formwork, only Props with dropheads remaining

Beam Formwork (UZ)

•With the UZ beam and bracket, beams are continously formed.
•No ties are required for beams of depth up to 80 cm.

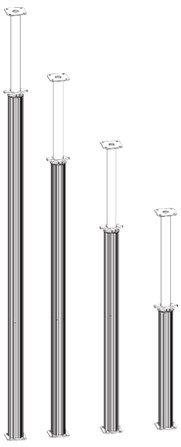
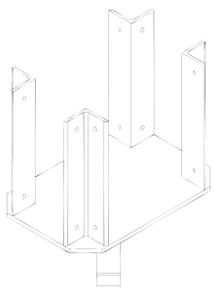
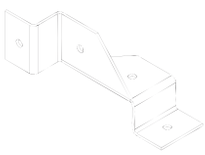




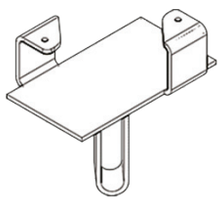
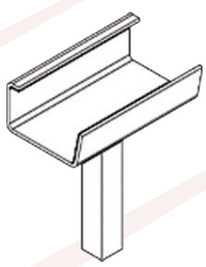
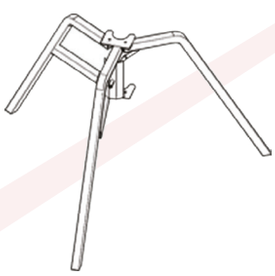
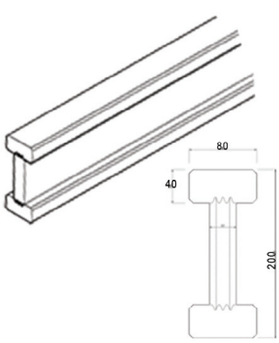
Permissible width (m) of influence of the bracket							
		Beam depth (m)					
		0.30	0.4	0.5	0.6	0.7	0.8
Slab thickness (m)	0	4.00	3.50	3.00	2.75	2.25	1.65
	0.2	3.64	2.85	2.50	1.80	1.35	1.05
	0.25	3.45	2.70	2.25	1.65	1.25	0.95
	0.3	3.30	2.40	2.00	1.50	1.15	0.90
	0.35	3.20	2.15	1.75	1.35	1.05	0.80

Width of influence of the bracket (m)
Beam depth (m)
Slab thickness (m)

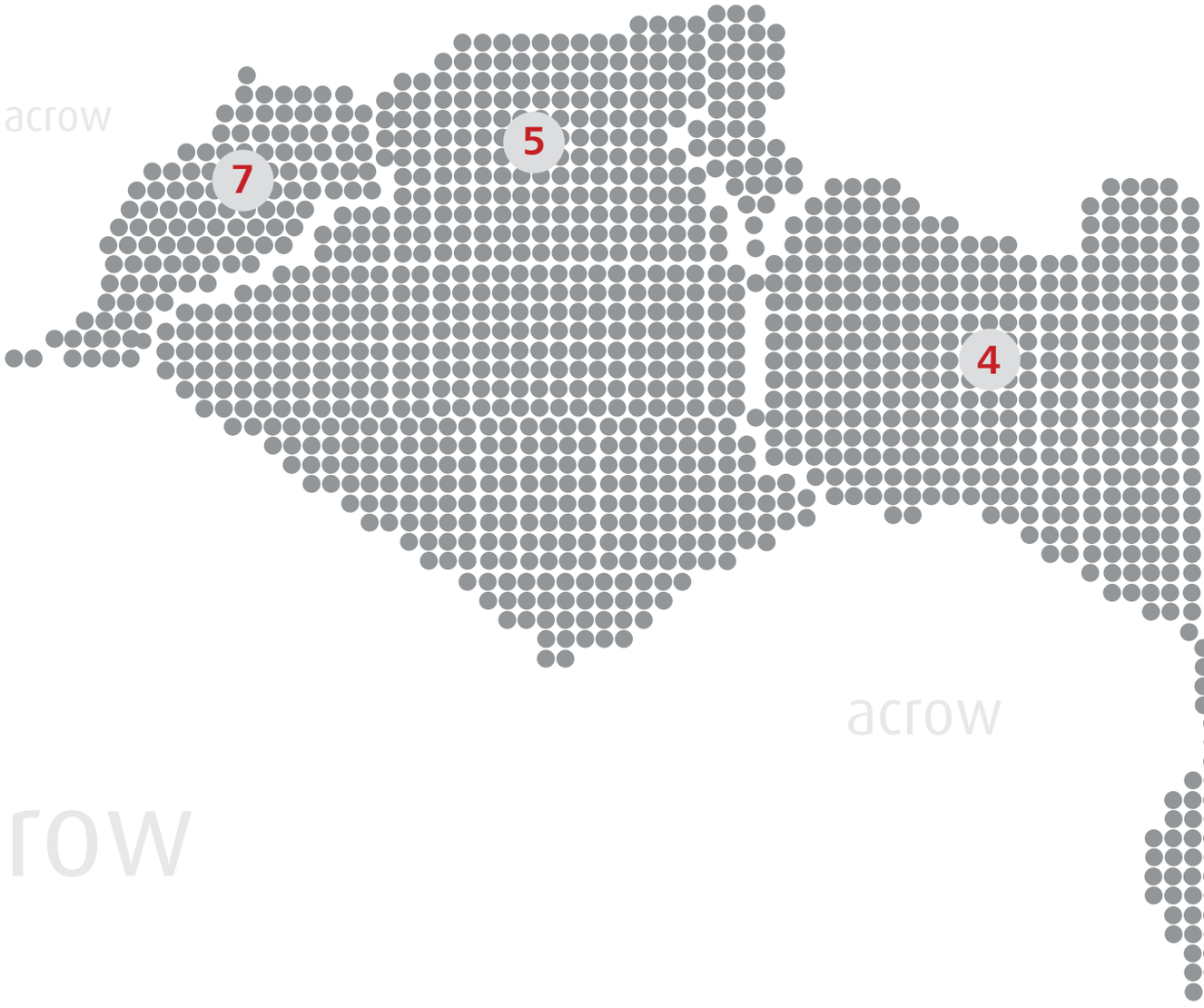
Components

Item	Description	Code no.	Weight (kg)
	Acrow Prop E30 Acrow Prop E35 Acrow Prop D40 Acrow Prop D45	6221177013381 6221177013392 6221177013403 6221177049744	16.90 20.50 22.00 25.40
	Alu Prop 300 Alu Prop 400 Alu Prop 500 Alu Prop 600	6221177080891 6221177080907 6221177080914 6221177080921	16.00 19.54 23.12 26.65
	Stirrup Head	6221177048440	4.40
	H20 to H20 connection	6221177006334	0.15

Components

Item	Description	Code no.	Weight (kg)
	Claw head (single) Claw head (double)	6221177013752 6221177013769	1.10 1.30
	Prop hanger (single) Prop hanger (double)	6221177013578 6221177013585	1.20 1.60
	Prop leg	6221177013776	12.00
	Timber H20 L=2.50 m Timber H20 L=2.90 m Timber H20 L=3.30 m Timber H20 L=3.60 m Timber H20 L=3.90 m Timber H20 L=4.50 m Timber H20 L=4.90 m Timber H20 L=5.90 m	6221177024062 6221177024076 6221177024093 6221177024109 6221177024116 6221177024130 6221177024147 6221177024154	12.50 14.50 16.50 18.00 19.50 22.50 24.50 29.50

Permissible bending Moment = 5.0 KNm
Permissible shear for = 11.0 KN
Approx. weight = 5.0 Kg/m



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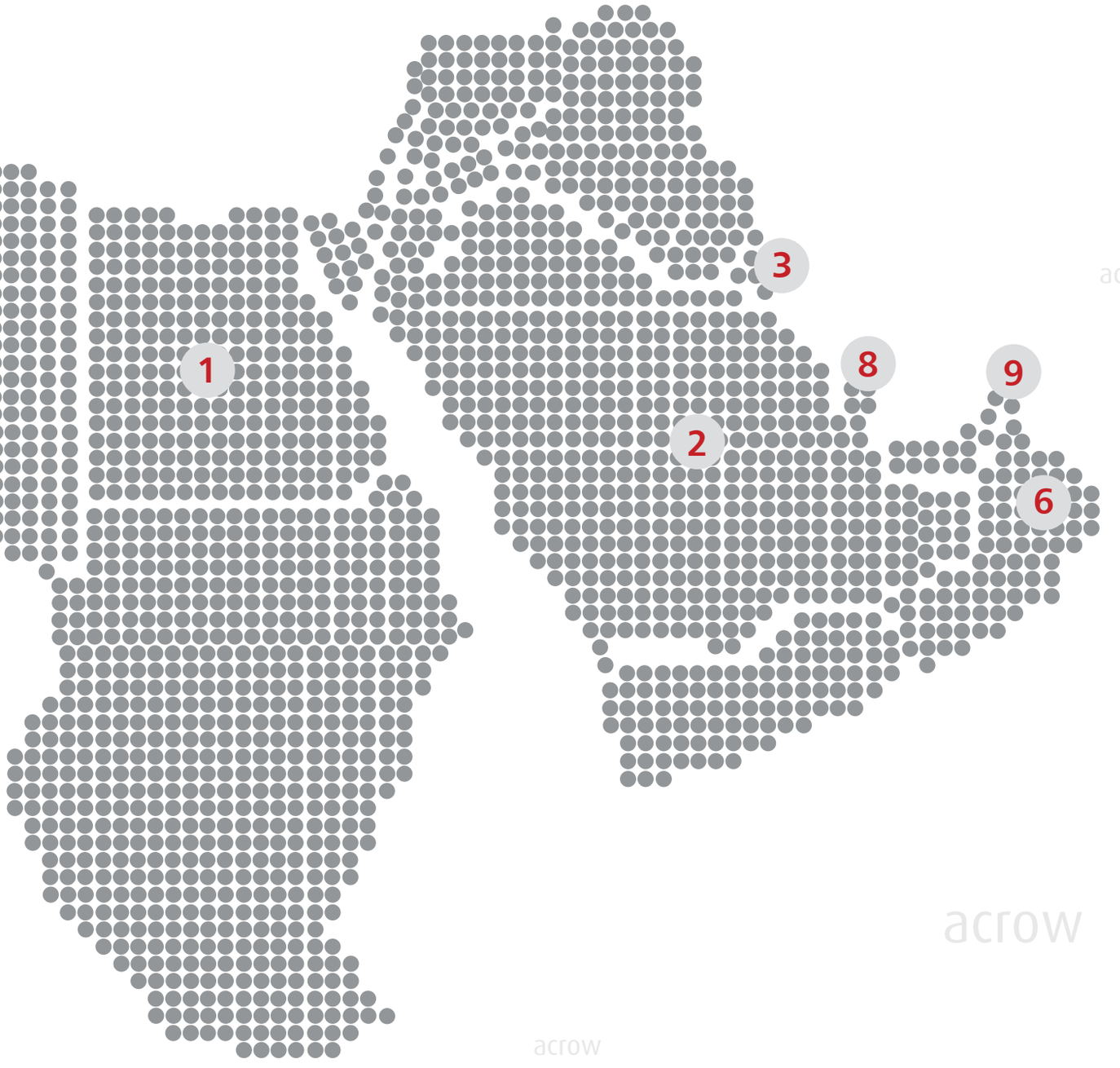
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ACROW
Formwork Technology


Acrow H20 & solider


Acrow One shutter solider


Acrow V.M.C


Acrow Tech-Form


Acrow Cuplock


Acrow Bridge table form

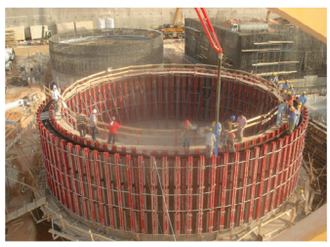

Acrow Circular column

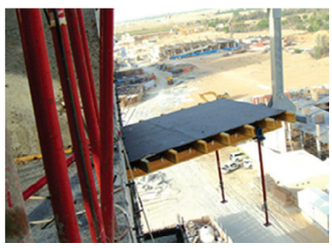

Acrow climbing system


Acrow shorbrace Table form


Acrow Big steel panel


Acrow European prop


Acrow Vari form


Acrow Prop table form


Acrow frame


Acrow early striking system


Acrow Ring lock

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